Chapter

1

Developments in Microbiology

Learning Objectives

After studying this chapter the students will be able to,

- *Know the microbes in space*
- *Understand the emerging microbes*
- *Learn about genetically modified microbes*
- *Appreciate nanoparticles production using microbes*
- *Know the important of microbiome*
- *Understand the applications of automated machines in microbial identification*

Chapter Outline

1.1 Microbes in Space

1.2 Emerging Microbes

1.3 Immunology

1.4 Molecular Biology and Genetic Engineering

1.5 Nanoparticles Production Using Microbes

1.6 Equipments

The field of microbiology, critical to human beings, not only due to the infectious diseases caused by these microbes but because "good" microorganisms are necessary to live on the planet and their extraordinary diversity of structure, function, habitat and applications are of paramount importance. Microorganisms (or microbes) inhabiting every corner of the globe, are indispensable to life on Earth and are responsible for some of the deadliest human diseases and form the basis of many industrial processes. This field of study could be considered as one of the most important areas of knowledge, considering that the bacteria in and on our bodies outnumber our own cells.

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Microbiology, an organismal discipline concerned with the properties of small forms of life or microorganisms. Bacteria neatly fit this definition, but what about fungi and algae? These two groups each contain members that are far from microscopic. On the other hand, certain animals, such as nematode worms, can be microscopic, yet are not considered to be the domain of the microbiologist. Viruses represent another special case; they are most certainly microscopic (indeed, most are submicroscopic), but by most accepted definitions they are not living. The concept of microbiology is remarkably broad in covering bacteria, protozoa and

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